

Submission LINALG1-PS1-SUB01

Question LINALG1-PS1-Q01

scratch: show step

1. $\det=6(5)-1(4)=26$
 2. Eliminate y , yielding $x=1$; substitute $\Rightarrow y=2$
 3. checks: $6(1)+1(2)=8$, $4(1)+5(2)=14$
- $\Rightarrow$ FINAL: $x=1$, $y=2$ (check units/interpretation)

Question LINALG1-PS1-Q02

scratch: show step

1. $\det(A)=2\cdot 3-2\cdot 2=2$
 2. $\det(A) \neq 0$, so A is invertible.
- $\Rightarrow$ FINAL: $\det=2$; invertible (check units/interpretation)

Question LINALG1-PS1-Q03

scratch: show step

1. $T(1,-1)=(1-2,3+1)=(-1,4)$
 2. Matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$
 3. Both coordinate rules have no constant term, so T is linear.
- $\Rightarrow$ FINAL: $T(1,-1)=(-1,4)$; matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$; linear (check units/interpretation)

Question LINALG1-PS1-Q04

scratch: show step

1. Set $c\blacksquare(1,0)+c\blacksquare(0,1)=(0,0)$.
 2. Coordinates give $c\blacksquare=0$ and $c\blacksquare=0$ only, so the set is independent.
- $\Rightarrow$ FINAL: only trivial combination; $\{(1,0),(0,1)\}$ is linearly independent (check units/interpretation)

Question LINALG1-PS1-Q05

scratch: show step

1. $\text{Null}(A)$ is the set of vectors sent to the zero vector.
 2. It is the solution space of $Ax=0$ and is a subspace.
- $\Rightarrow$ FINAL: the vectors mapped to zero; exactly the solution space of $Ax=0$ (check units/interpretation)

Question LINALG1-PS1-Q06

scratch: show step

1. $\det=5(3)-2(3)=9$
 2. Eliminate y , yielding $x=1$; substitute $\Rightarrow y=2$
 3. checks: $5(1)+2(2)=9$, $3(1)+3(2)=9$
- $\Rightarrow$ FINAL: $x=1$, $y=2$ (check units/interpretation)